Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Dsaise
Project / Solution Name	HalaTuju
Name (Leader)	Leow Shen En
Email & Phone (Leader)	leowshenen@gmail.com | 018-2335269
University Affiliation(s)	Universiti Malaya
Live Prototype URL	https://halatuju.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/shenenleow10521/SdcHackathon

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

A JavaScript browser app displays a 3D career map served by Node 22, TypeScript and Express. CV text, PDFs and scans enter the API; local Tesseract OCR reads scans and a scrubber removes personal details before AI calls. Gemini extracts CV evidence and reviews market claims. BAAI/bge-m3 on RunPod Serverless matches CV phrases against 13,939 ESCO skills; a team-trained ridge model estimates task exposure from those embeddings. PostgreSQL stores profiles, sourced knowledge versions and saved plans. Fixed formulas in the shared engine calculate matches, routes and 1,000-future stress tests. A daily researcher obtains dated material from vetted sources through Tavily, RSS and public research APIs. Quotations are checked, a second model reviews claims, and fixed triage sends route-changing updates to a person for approval. The live Replit URL is https://halatuju.replit.app; GitHub Actions runs type-checks and tests. Sign-in uses Replit-managed Clerk; guests can use the app without an account.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Node 22; TypeScript 7; Express 5.3	Server and HTTP API	MIT / Apache-2.0
zod 4.6; multer 2.5; unpdf 1.9	Validation, uploads, PDF extraction	MIT / Apache-2.0
tesseract.js 7; eng and msa models	Local OCR	Apache-2.0
pg 8.23; PGlite 0.5	PostgreSQL access and local tests	MIT / Apache-2.0
fast-xml-parser 5.11; dotenv 18	Feed parsing and local config	MIT / BSD-2-Clause
3d-force-graph 1.79; three.js; GSAP 3.12	3D map and animation	MIT / GreenSock standard

Name & Version	Purpose in Project	License
Lucide; Geist font	Icons and typography	ISC / SIL OFL 1.1
@clerk/express 2.1; @clerk/clerk-js 6.40; http-proxy-middleware 4.2	Sign-in and sessions (Replit-managed Clerk)	MIT
Vitest 5; supertest 7	418 automated tests	MIT
Python 3.11; PyTorch; sentence-transformers; scikit-learn; NumPy; pandas; SciPy	Model training and evaluation	Python / BSD / Apache-2.0 as applicable

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API	CV parsing, signal review, explanations; embedding fallback	Free tier
RunPod Serverless API a7ipau1nmqogxs	BAAI/bge-m3 embeddings	Sponsor credit
RunPod GPU Cloud API	Launch and monitor final training pod	Sponsor credit
Tavily Search API	Search vetted domains for market signals	Free tier
ESCO REST API; data.gov.my/OpenDOSM API	Build-time taxonomy and wage data	Free
arXiv; OpenAlex; GDELT; Hacker News Algolia; RSS	Public research and news signals	Free
Clerk (via Replit)	Sign-in and sessions	Sponsor credit

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Reserved VM and PostgreSQL 16	Live app at halatuju.replit.app; its only database (Replit PostgreSQL 16)	Sponsor platform
RunPod GPU Cloud	Final ridge-model training on RTX 4090 pod 38jijj8fe5dazu	Sponsor credit; Secure Cloud
RunPod Serverless	bge-m3 runtime embeddings, endpoint a7ipau1nmqogxs	Sponsor credit; 0–1 workers

Service & Provider	Purpose in Project	Plan / Region
Supabase	PostgreSQL during local development only	Free tier
GitHub and Actions	Repository, type-check and test CI	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gemini-3.5-flash-lite	Google Gemini API	Pre-trained; extraction and explanation	Gemini API terms
gemini-3.1-flash-lite	Google Gemini API	Pre-trained; review and fallback	Gemini API terms
gemini-embedding-001	Google Gemini API	Pre-trained; fallback embeddings	Gemini API terms
BAAI/bge-m3	BAAI / Hugging Face; RunPod	Pre-trained; 1,024-d embeddings	MIT
HalaTuju ridge exposure estimator	Team; trained on RunPod	Self-trained on human task labels	Project Apache-2.0; labels MIT
Tesseract eng and msa LSTM	Tesseract OCR	Pre-trained; local OCR	Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ESCO v1.2.1; ESCO–O*NET crosswalk	esco.ec.europa.eu	Public taxonomy/mapping	EC reuse with attribution	N
O*NET tasks and ratings	onetcenter.org	Public occupational data	CC BY 4.0	N
GPTs are GPTs task labels	github.com/openai/GPTs-are-GPTs	Public training/evaluation	MIT	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ILO WP140; Anthropic Economic Index	ilo.org; huggingface.co/Anthropic/EconomicIndex	Public research	CC BY 4.0 / CC BY	N
DOSM/OpenDOSM; MyCOL 2024/25	data.gov.my; TalentCorp	Public labour data	CC BY 4.0 / facts cited	N
Public course listings and dated news/research	Providers; vetted websites	Public facts, links and short quotes	Source-specific; links cited	N after scrub
50 CV benchmark cases; demo persona; sample catalogue	Created for project	Synthetic	Project Apache-2.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic; Opus models)	Code, tests, debugging and documentation	Major
Claude agents (Anthropic)	Synthetic CVs, blind answer keys and judging	Moderate
Replit Agent (Replit)	migrated sign-in to Replit-managed Clerk	Moderate
Google Gemini gemini-3.5-flash	A discarded first benchmark judge run	Minor
faster-whisper large-v3	Local transcription of organiser tutorial	Minor
OpenAI Codex (GPT-6; this document task)	Review project and complete these declarations	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team directed the design and checked AI-assisted code with a TypeScript type-check and 418 automated tests. Held-out synthetic CV evaluations, a blind privacy run and stored model metrics exposed weaknesses that are reported in the repository. Source quotations are checked before use; route-changing knowledge updates need human approval and can be undone. The team remains responsible for all submitted claims, licences and links.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
RunPod	US$25 GPU/Serverless credit	RTX 4090 training run of our AI-exposure model and runtime bge-m3 embeddings; pod 38jijj8fe5dazu, endpoint a7ipau1nmqogxs
Replit	Core platform, Reserved VM and PostgreSQL	Live prototype at halatuju.replit.app; Replit PostgreSQL 16 holds the knowledge base; all keys in Replit Secrets. Proof: SDC2026_Dsaise_Replit_POU.pdf

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

RunPod provides the GPU training run and matching embeddings. A local bge-m3 server, Gemini embeddings, then labelled keyword matching are fallback paths. Replit hosts the live app; the Node 22 and PostgreSQL app can run on a comparable host.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The app stores confirmed skills and levels, current role, study hours, goals and saved plans under a profile ID. Sign-in uses Replit-managed Clerk; guests can upload a CV without an account, tied to a signed browser cookie. CV files are read in memory and not stored. Scans use local OCR; names, identifiers, contact details and addresses are scrubbed before CV text is sent to Gemini or RunPod. Held-out synthetic tests found 99.4% removal after fixes, which is not a guarantee for real CVs. Accounts and guests can delete their own profile data. Keys belong in Replit Secrets; staff routes require role checks or an admin token. Retention of saved profile data lasts until the user deletes it; any provider-side retention follows provider terms.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

A misread article could alter advice; trusted domains, checked quotes, a second reviewer, fixed triage, human approval for route changes and version rollback reduce that risk. Our exposure model has higher per-task MAE than a nearest-neighbour baseline, so exposure remains an estimate and its measured error informs stress tests. The 16-job route catalogue and synthetic CV benchmark do not establish real-world outcomes; samples are labelled. Gemini quotas and RunPod cold starts can delay calls; fallbacks are labelled. CV parsing and PII removal can fail on unfamiliar documents, requiring user review. The emerging-skill watchlist is not treated as verified demand.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



24/10/2026

Appendix A: System block diagram

